



Staff Software Engineer, AI

CONTACT

- New York, USA
- rajani.maski@gmail.com
- (516) 309-0398
- linkedin.com/in/rajanimaski

SPEAKING

Optimized AI Conference 2026

Agentic Search Patterns **SPEAKER**

OpenSearchCon India

Generative Discovery: Intent, Context and Cognition **SPEAKER**

CERTIFICATIONS

AWS Certified GenAI Developer Professional

DeepLearning.AI

40+ courses — RAG, Agents, LLMOps, Transformers

EDUCATION

M.S. Computer Science

San Francisco State University

B.E. Engineering

Visvesvaraya Technological University

CORE AI SKILLS

- Agentic Systems
- Multimodal AI
- Prompt Engineering
- RAG Pipelines
- Vector Search
- Embeddings
- LLMOps
- OpenSearch
- AWS Bedrock
- NVIDIA NIM
- NVIDIA NeMo
- NeMo Retriever
- Triton Inference
- cuVS / RAPIDS
- TensorRT-LLM

LANGUAGES & INFRA

- Python
- Java
- Scala
- Bash
- AWS
- Kubernetes
- Terraform

Rajani Maski

Staff Software Engineer, AI

PROFILE

Staff AI Engineer with 15+ years of experience from classical IR to modern AI, including model training, semantic search, hybrid retrieval, relevance ranking, learning-to-rank (LTR), and production retrieval systems serving 2000+ RPS. Brings hands-on experience working directly on customer projects as a Senior Consulting Engineer, and now focuses on Generative Discovery, building agentic and multimodal systems across retrieval, ranking, and image generation. Operating at the intersection of retrieval, intelligence, and real-world customer impact.

EXPERIENCE

Staff Software Engineer, AI

Shutterstock · New York · May 2022 – Present · 4 yrs 1 mo

- Architecting **Generative Discovery infrastructure** powering RAG-style GenAI applications, combining dense vector retrieval, hybrid search, and metadata grounding at 2000+ RPS across hundreds of nodes.
- Drove **25% latency and cost reduction** through retrieval architecture optimization while simultaneously improving relevance metrics, decoupling scale from quality tradeoffs.
- Building **multimodal embedding pipelines** supporting image + text semantic search, with MLflow and SageMaker for feature generation and model lifecycle management.
- Continuously improving the **ranking pipeline** to retrieve and rerank results for the best candidate match, leveraging Learning-to-Rank (LTR), bi-encoders, cross-encoders, and related techniques.
- Working with the team to automate **asset metadata tag generation** pipelines for images and videos, improving search accuracy and retrieval coverage.
- Designed **offline and online evaluation frameworks** for retrieval quality using precision/recall, relevance judgments, and human-in-the-loop feedback loops.
- Led **Solr to OpenSearch migration**, redesigning the retrieval layer for cloud-native scale and enabling ML-powered ranking pipelines including Learning-to-Rank.

Professional Services Consulting Engineer

Redis · New York · Dec 2020 – May 2022 · 1 yr 5 mos

- Designed and deployed **vector similarity search systems** using Redis as a senior consulting engineer, advising enterprise customers and translating business discovery needs into production retrieval architectures and reusable reference designs.
- Delivered projects using **RediSearch and RedisGraph** for financial banking customers, building knowledge graphs for CDC of banking transactions.

Senior Search Consultant

Lucidworks · New York · Jun 2018 – Dec 2020 · 2 yrs 6 mos

- Architected enterprise search platforms using **Lucidworks Fusion, Solr and Spark**; implemented collaborative recommendations, personalized search, and signal-based boosting with ML-based relevance pipelines.
- Led customer-facing relevance engagements end-to-end — from discovery and signal modeling to LTR deployment — measurably improving search quality for enterprise clients.

Prior Experience (2009–2017)

Search Consultant · Lucidworks India | Senior Software Engineer · Target.com India | Sr. Java Programmer · Happiest Minds | Search Consultant · Sony India | Software Engineer · PointCross Life Sciences | Search Intern · Glassdoor USA